NEOS — New Earth Operating System

What is NEOS?

NEOS is a governance skill stack — a complete, modular protocol for how groups of humans coordinate without a centralized authority telling them what to do.

Think of it as an operating system for organizations that refuse to have a boss. Not because bosses are bad people, but because concentrating authority in any single point — a person, a board, a token — is a structural failure mode that eventually corrupts the system. Every time. Michels called it the Iron Law of Oligarchy. NEOS is an engineering response to that law.

Each "skill" in the stack is a self-contained process document: when this situation arises, here is what happens, who has authority, what the boundaries are, what the output looks like, and what happens when it breaks. An AI agent can read a single skill and guide a participant through the process. An ecosystem architect can fork the stack and configure it for their community.

The first ecosystem running on NEOS is **OmniOne**, a non-sovereign coordination network stewarded by Green Earth Vision (GEV), operating out of Bali.

What the Skill Stack Can Do

When fully built, NEOS gives an ecosystem the structural capacity to:

Make Binding Agreements Without a Legal Department

Create, amend, review, and sunset agreements at every level — from a shared kitchen's quiet hours policy to a universal commitment field that every participant consents to upon entry. Every agreement is versioned, registered, and traceable. No agreement exists by implication; everything is written down.

Make Decisions Without Voting

The ACT protocol (Advice, Consent, Test) replaces majority-rule voting with a three-phase process. First, the proposer gathers advice from everyone affected. Then, the group checks for reasoned objections — not enthusiasm, just the absence of "this would harm our shared aim." Then, the decision is implemented on a reversible, time-limited test basis. If it doesn't work, it reverts. Six levels of escalation (the GAIA model) handle cases where a proposal stalls.

Define Authority Without Creating Hierarchy

Authority in NEOS is scoped, temporary, and reviewable. A role has a defined domain, a defined expiry, and defined boundaries it cannot cross. Domain mapping uses the Sociocracy 3.0 eleven-element contract. Roles are assigned, transferred, reviewed, and sunset through structural processes — not through a manager's judgment call.

Coordinate Economics Without Capital Controlling Governance

Resource requests, funding pool stewardship, and participatory allocation are governed by Ostrom's eight commons principles. The structural rule: economic contribution does not purchase governance authority. A donor's million-dollar gift cannot buy a seat at the decision table. Influence currencies (Current-Sees in OmniOne) are distributed equally — 111 per person.

Coordinate Across Autonomous Units Without a Central Office

Cross-unit requests, shared resource stewardship, federation agreements, and polycentric conflict navigation allow autonomous zones (AZPOs) to cooperate without subordination. The design pattern draws from Ostrom's polycentric governance and the Ottoman millet system's structural separation — minus the Sultan.

Resolve Conflict Without Courts or Punishment

Harm circles, NVC dialogue, repair agreements, coaching interventions, and escalation triage provide a full conflict resolution pathway grounded in Transformative Justice and Restorative Justice. The goal is repair, not punishment. The structural guarantee: accessible, low-barrier conflict resolution at every level.

Detect and Resist Capture Before It's Too Late

Continuous governance health auditing monitors for four capture types: capital capture (a funder gains disproportionate influence), charisma capture (a personality becomes too costly to challenge), emergency capture (temporary authority becomes permanent), and ossification capture

(leadership calcifies despite formal rotation). These are measured with indicators, not hunted with suspicion.

Handle Emergencies Without Losing Democracy

A circuit breaker pattern borrowed from distributed systems engineering provides three states: Normal, Crisis, and Recovery. Emergency authority expands within pre-authorized scopes. Auto-reversion timers start the moment a crisis is declared. The Recovery state cannot be skipped. Agamben's warning — that emergency powers are how democracies die — is treated as an engineering constraint, not a philosophical concern.

Remember and Learn From Past Decisions

Decision records, semantic tagging, precedent search, agreement versioning, and precedent challenge create a governance memory system. Past decisions can be found, understood, and challenged if circumstances change. The design draws from legal stare decisis: precedent guides but does not bind.

Let People Leave Without Punishment or Data Loss

Exit is a structural right, not a failure. Voluntary departure follows an orderly process: commitments are unwound, governance history is exported in a portable format (GDPR Article 20 principles), and the ecosystem degrades gracefully. No single departure — not even the most active or connected member — should cause governance to collapse. If it would, that's a capture indicator.

The Unified Framework

Ten Layers, One Stack

NEOS is organized into ten skill layers. Each layer addresses a distinct governance domain. Together, they form a complete coordination protocol.

```
Layer I    Agreement           What we've committed to
Layer II   Authority & Role    Who can do what, within what boundaries
Layer III  Decision (ACT)      How we change what we've committed to
Layer IV   Economic Coord.     How resources flow without capital controlling governance
Layer V    Inter-Unit Coord.   How autonomous units cooperate without subordination
Layer VI   Conflict & Repair   How we repair harm without punishment
Layer VII  Safeguard & Capture How we detect governance degradation before it's too late
Layer VIII Emergency Handling   How we act fast without losing democracy permanently
Layer IX   Memory & Trace      How we remember and learn from past decisions
Layer X    Exit & Portability  How people leave freely with their data and dignity
```

The Agreement Hierarchy

All governance in NEOS flows from explicit, written agreements organized in a strict hierarchy. Higher-level agreements override lower-level ones. Conflicts resolve upward.

```
Universal Agreement Field (UAF)
  └── Ecosystem Agreement (e.g., Master Plan)
        └── Access Agreement (e.g., space-specific rules)
              └── Stewardship Agreement (e.g., role-specific commitments)
                    └── AZPO Agreement Field (organizational unit agreements)
                          └── Culture Code (circle-internal norms)
                                └── Personal Commitments
```

Ten Non-Negotiable Structural Principles

These are not values to aspire to. They are engineering constraints that every skill must satisfy.

1. **No Sovereign Authority** — No person, council, or token holds permanent override power.
2. **Scoped Authority** — All authority exists within defined domains and must expire or be reviewable.
3. **Voluntary Participation** — Exit is always structurally possible. Consent under duress is not consent.
4. **Capital Does Not Equal Power** — Economic contribution cannot purchase governance authority.
5. **Solutionary Decision-Making** — Advice, Consent, Test. Not voting. Not executive decree.
6. **Local Failure Containment** — One unit's failure does not cascade to others.
7. **Explicit Agreements** — Nothing governs by implication. If it's not written down, it's not binding.

8. **Capture Resistance** — The system structurally resists concentration of power through capital, charisma, emergency, or inertia.
9. **Pluralism** — Multiple value systems coexist through shared agreements, not shared beliefs.
10. **Emergency Contraction** — Temporary authority expansion auto-expires. The crisis state cannot become the new normal.

The Stress-Test Protocol

Every skill in the stack is tested against seven scenarios designed to break governance systems:

Scenario	What It Tests
Capital Influx	A major donor offers funding contingent on governance concessions. Does capital buy influence?
Emergency Crisis	A natural disaster or acute threat requires fast decisions. Does emergency authority become permanent?
Leadership Charisma Capture	A beloved leader's proposals pass without scrutiny. Can the system challenge popularity?
High Conflict / Polarization	Two factions want incompatible things. Can the system find third solutions or gracefully partition?
Large-Scale Replication	The ecosystem grows from 50 to 5,000 members. Do processes scale or break?
External Legal Pressure	A government demands compliance with a mandate that contradicts NEOS principles. Where is the sovereignty boundary?
Sudden Exit of 30%	Nearly a third of participants leave at once. Does governance degrade gracefully or collapse?

A skill that cannot survive all seven scenarios is not ready for deployment.

How Skills Work

Each skill is a markdown document with YAML frontmatter, structured into twelve required sections:

Section	Purpose
A. Structural Problem It Solves	Why this skill exists — what fails without it
B. Domain Scope	Where this skill applies
C. Trigger Conditions	What activates this skill
D. Required Inputs	What information the skill needs to run
E. Step-by-Step Process	The actual procedure — actionable, not philosophical
F. Output Artifact	What gets produced (document, record, decision)
G. Authority Boundary Check	Who has authority and where the limits are
H. Capture Resistance Check	How this skill resists the four capture types
I. Failure Containment Logic	What happens when the process breaks
J. Expiry / Review Condition	When this skill's outputs expire or get reviewed
K. Exit Compatibility Check	What happens to this skill's outputs when someone leaves
L. Cross-Unit Interoperability	How this skill works across autonomous units

Every skill includes a concrete OmniOne walkthrough showing the process with real roles, real scenarios, and at least one complication. Every skill includes full narrative stress-test results against all seven scenarios.

An AI agent reads a skill and can guide a participant through it. An ecosystem architect reads a skill and can adapt it for their community.

The Full Skill Inventory

54 governance skills across 10 layers:

Layer I: Agreement (5 skills)

- `agreement-creation` — Create a new binding agreement through consent-based process
- `agreement-amendment` — Modify an existing agreement with traceability

- `agreement-review` — Periodic re-validation of agreements against current conditions
- `agreement-registry` — Single source of truth for all active agreements
- `universal-agreement-field` — The root agreement every participant consents to upon entry

Layer II: Authority & Role (7 skills)

- `domain-mapping` — Define domains using S3 eleven-element contract
- `member-lifecycle` — Onboarding, status changes, and offboarding
- `role-assignment` — Assign roles through consent-based process
- `authority-boundary-negotiation` — Negotiate and resolve boundary disputes
- `role-transfer` — Hand off roles with continuity
- `domain-review` — Periodic review of domain definitions
- `role-sunset` — Graceful retirement of roles no longer needed

Layer III: Decision / ACT Engine (6 skills)

- `proposal-creation` — Draft and submit proposals with synergy check
- `act-advice-phase` — Gather input from all impacted parties
- `act-consent-phase` — Check for reasoned objections, integrate them
- `act-test-phase` — Implement decisions reversibly with defined review
- `proposal-resolution` — GAIA 6-level escalation for stalled proposals
- `consensus-check` — Verify whether a group has reached agreement

Layer IV: Economic Coordination (5 skills)

- `resource-request` — Request resources through governed process
- `funding-pool-stewardship` — Manage shared funding pools with Ostrom principles
- `participatory-allocation` — Collective resource allocation decisions
- `commons-monitoring` — Monitor shared resource health
- `access-economy-transition` — Transition from ownership to access models

Layer V: Inter-Unit Coordination (5 skills)

- `cross-azpo-request` — Request cooperation across autonomous units
- `shared-resource-stewardship` — Manage resources shared between units

- `federation-agreement` — Form agreements between autonomous units
- `inter-unit-liaison` — Define liaison roles between units
- `polycentric-conflict-navigation` — Navigate conflicts across unit boundaries

Layer VI: Conflict & Repair (6 skills)

- `harm-circle` — Convene affected parties for restorative process
- `nvc-dialogue` — Structured nonviolent communication process
- `escalation-triage` — Route conflicts to appropriate resolution level
- `repair-agreement` — Create binding repair commitments
- `coaching-intervention` — Third-party coaching for stuck conflicts
- `community-impact-assessment` — Assess ripple effects of conflicts and resolutions

Layer VII: Safeguard & Capture Detection (5 skills)

- `governance-health-audit` — Measure governance health with defined indicators
- `capture-pattern-recognition` — Match governance patterns against known capture signatures
- `safeguard-trigger-design` — Define thresholds that activate structural safeguards
- `independent-monitoring` — Ensure monitoring independence from those being monitored
- `structural-diversity-maintenance` — Proactively maintain governance diversity

Layer VIII: Emergency Handling (5 skills)

- `emergency-criteria-design` — Define objective criteria for emergency declaration
- `pre-authorization-protocol` — Pre-authorize emergency roles before crises occur
- `crisis-coordination` — Coordinate during active emergencies
- `emergency-reversion` — Structurally revert to normal governance after crisis
- `post-emergency-review` — Mandatory review of every emergency invocation

Layer IX: Memory & Traceability (5 skills)

- `decision-record` — Record any governance decision in a generic wrapper
- `semantic-tagging` — Tag decisions for findability across layers
- `precedent-search` — Search past decisions for relevant precedent
- `agreement-versioning` — Track the full version history of agreements

- `precedent-challenge` — Challenge or distinguish past precedent

Layer X: Exit & Portability (5 skills)

- `voluntary-exit` — Structured, dignified departure process
- `commitment-unwinding` — Orderly unwinding of active commitments
- `portable-record` — Export governance history in portable format
- `azpo-dissolution` — Dissolve an autonomous unit gracefully
- `re-entry-integration` — Return to the ecosystem after departure

Theoretical Foundations

NEOS does not invent governance theory. It synthesizes established frameworks into an engineering discipline:

Framework	What NEOS Uses From It
Sociocracy 3.0	Consent-based decision-making, domain model, circle structure
Ostrom's Commons Governance	Eight principles for managing shared resources, polycentric governance
Transformative Justice	Repair over punishment, community accountability
Nonviolent Communication (NVC)	Structured dialogue for conflict resolution
Michels' Iron Law of Oligarchy	The structural problem NEOS exists to resist
Agamben's State of Exception	The mechanism by which emergency powers become permanent
Hirschman's Exit, Voice, Loyalty	Exit as a governance right, not a failure
Circuit Breaker Pattern	Emergency state machine with auto-reversion
GDPR Article 20	Data portability as a governance right
Holacracy	Role-based authority, organizational clarity
Laloux Teal Organizations	Self-management, evolutionary purpose
Legal Stare Decisis	Precedent-based governance memory

Architecture

```
neos-core/
├── layer-01-agreement/        5 skills
├── layer-02-authority-role/    7 skills
├── layer-03-act-engine/        6 skills
├── layer-04-economic-coord/    5 skills
├── layer-05-inter-unit/        5 skills
├── layer-06-conflict-repair/   6 skills
├── layer-07-safeguard/         5 skills
├── layer-08-emergency/         5 skills
├── layer-09-memory-trace/      5 skills
├── layer-10-exit-portability/  5 skills
├── NEOS_PRINCIPLES.md          The 10 non-negotiable structural principles
├── STRESS_TEST_PROTOCOL.md     The 7 scenarios and how to apply them
└── README.md                   Orientation for all three audiences

scripts/
├── validate_skill.py           Validates any SKILL.md against the format spec
├── stress_test_report.py       Aggregates stress-test results across all skills
└── package_zip.py              Packages the stack for offline distribution

reference-docs/                 Source material and research
```

Every skill directory contains: - `SKILL.md` — The skill document (under 500 lines) - `assets/` — Templates and schemas (YAML) - `references/` — Supporting material loaded as needed

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Who This Is For

AI Agents read individual skills and guide participants through governance processes. The YAML frontmatter, consistent section structure, and cross-references by skill name are designed for machine consumption.

Ecosystem Architects fork `neos-core/`, replace OmniOne example blocks with their own community's specifics, and run `validate_skill.py` to confirm their configuration is structurally sound.

Curious Observers read this document and the layer READMEs to understand what NEOS is, what it is not, and whether it addresses governance problems they recognize.

NEOS is not a utopia. It is an engineering specification for governance systems that resist their own failure modes. Every mechanism in the stack exists because there is a documented historical pattern of governance systems failing in exactly that way. The goal is not perfection — it is structural resistance to the specific, predictable ways that coordination breaks down.